

Min Namgung

PH.D. STUDENT

Kenneth H. Keller Hall, 200 Union St SE, Minneapolis, MN 55455

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Research Interest

Multimodal Region Representation Learning, Spatial AI, Data Mining

Education

University of Minnesota – Twin Cities

PH.D. student Computer Science and Engineering

- Advisor: Prof. Yao-Yi Chiang

Minneapolis, MN

Aug 2021 – Present

Purdue University Fort Wayne

M.Sc. Computer Science

- Advisor: Prof. Jin Soung Yoo
- Thesis: Performance Comparison of Public Bike Demand Predictions: The Impact of Weather and Air Pollution
- Received graduate assistant scholarship (1-year)

Fort Wayne, IN

Aug 2019 - Dec 2020

Purdue University Fort Wayne

B.Sc. Computer Science

- Advisor: Prof. Jin Soung Yoo
- Received chancellor's merit award scholarship (2-year)

Fort Wayne, IN

May 2019

Honors & Awards

2025	The Best use of map U-Spatial Mapping Prize	Minneapolis, MN
2022-24	Student Travel Award , ACM SIGSPATIAL	USA
2023	First Place, Best Paper Award , 2023 UCGIS	New Haven, CT, USA
2023	Student Travel Award , SIAM International Conference on Data Mining	Minneapolis, MN
2023	Best PC member (Reviewer) , The 32nd ACM CIKM	Birmingham, UK
2019	Graduate Assistant Scholarship (1-year) , Purdue University Fort Wayne	Fort Wayne, IN
2017	Chancellor's Merit Award Scholarship (2-year) , Purdue University Fort Wayne	Fort Wayne, IN
2019	Semester Honors' List , Purdue University Fort Wayne	Fort Wayne, IN
2018-19	Dean's List , Purdue University Fort Wayne	Fort Wayne, IN

Publications

- [C7] **Namgung, M.**, Chiang, Y.-Y., Omitaomu, F., (2025). Context-aware Multimodal Region Representation Learning for Urban Health Risk Prediction, ***Under submission to UrbanAI workshop at ACM SIGSPATIAL' 25***
- [C6] **Namgung, M.**, Lee, JH., Ding, F., Chiang, Y.-Y. (2025). Transit for All: Mapping Equitable Bike2Subway Connection using Region Representation Learning. arXiv preprint arXiv:2506.15113., ***Accepted to ACM SIGSPATIAL' 25***
- [C5] **Namgung, M.**, Lee, JH., Lin, Y., Chiang, Y.-Y. (2025). Capturing Shared and Unique Information in Multimodal Region Representations for Urban Mobility Prediction, ***Accepted to Urban Mobility Foundation Models workshop at ACM SIGSPATIAL' 25***
- [C4] **Namgung, M.**, Lin, Y., Lee, JH., Chiang, Y.-Y. (2025) Less is More: Multimodal Region Representation via Pairwise Inter-view Learning., arXiv preprint arXiv:2505.18178
- [C3] Jeong, M., **Namgung, M.**, Kim, ZM., Kang, D., Chiang, Y.-Y., Hero, A., (2025) Anchors Aweigh! Sail for Optimal Unified Multi-Modal Representations. arXiv preprint arXiv:2410.02086
- [C2] Kim, J., Li, Z., Lin, Y., **Namgung, M.**, Jang, L., Chiang, Y.-Y. (2023) The mapKurator System: A Complete Pipeline for Extracting and Linking Text from Historical Maps. In *Proceedings of the 31st ACM SIGSPATIAL International Conference on Advances in Geographic Information Systems*, Article 35, 1–4, Hamburg, Germany.
- [B1] Chiang, Y.-Y., Chen, M., Duan, W., Kim, J., Knoblock, C. A., Leyk, S., Li, Z., Lin, Y., **Namgung, M.**, Shbita, B., Uhl, J. H. (2023). *GeoAI for the Digitization of Historical Maps*. In *Handbook of Geospatial Artificial Intelligence* (pp. 217-247). CRC Press.

- [C1] **Namgung, M.** and Chiang, Y.-Y. (2022). Incorporating spatial context for post-OCR in map images. In *Proceedings of the 5th ACM SIGSPATIAL International Workshop on AI for Geographic Knowledge Discovery*, pp. 14-17, Seattle, WA, USA.

Posters

- [P1] **M. Namgung**, Chiang, Y.-Y. (2022) Incorporating Prior Knowledge to Forecast Fine-Grained Cloud-Top Temperature. *The 2022 University Consortium for Geographic Information Science Symposium*, Syracuse, NY, USA

Invited Talks

- [T5] **Namgung, M.** (October 2024), “Multimodal Learning in Geospatial Application”, *Electronics and Telecommunications Research Institute (ETRI)*, Pangyo, South Korea
- [T4] Ailshire, J., Chiang, Y.-Y., **Namgung, M.**, Nichols, E., (August 2024), “Measuring neighborhood attributes: Machine learning approaches and development of an audit tool for assessment of model performance”, *LASI-DAD Project Meeting*, Banff, Canada
- [T3] **Namgung, M.**, Chen, T., Chiang, Y.-Y., Ailshire, J., (July 2023), “Multimodal Artificial Intelligence to Enhance Environmental Data Integration”, *LASI-DAD Project Meeting*, Amsterdam, Netherlands
- [T2] **Namgung, M.**, (June 2023), “Representation Learning of Regions using Unevenly Distributed, Incomplete Multi-Modal Data.”, *2023 UCGIS*, New Haven, CT, USA
- [T1] **Namgung, M.**, (February 2023), “A pathway to become computer science Ph.D. student”, *Thomas Jefferson High School for Science and Technology*, Alexandria, VA

Work Experience

2025-now.	Oak Ridge National Lab Research Intern (June-Dec)	Knoxville, TN
2024	Oak Ridge National Lab Research Intern (June-August)	Knoxville, TN
2018-19	Parkview Health , Back-end Developer (Capstone project)	Purdue Univ. FW
2018-19	Purdue Indiana Manufacturing Competitiveness Center , Software Engineer	Purdue Univ. FW

Teaching Experience

Fall 2024	CSCI5523: Intro to Data Mining , TA	Univ. of Minnesota
Fall 2022	CSCI4707: Practice of Database System , TA	Univ. of Minnesota
Sep 2022	CSCI1913: Introduction to Algorithms, Data Structures and Development , TA	Univ. of Minnesota
Fall 2021	CSCI1933: Introduction to Algorithms and Data Structures , TA	Univ. of Minnesota
Fall 2020	CS364: Introduction to Database Systems , TA	Purdue Univ. FW
Sep 2020	CS161: Introduction to Computer Science II , TA	Purdue Univ. FW
Fall 2019	CS160: Introduction to Computer Science I , TA	Purdue Univ. FW
May 2018	CS112: Computer Science for Everyone , Undergraduate TA (For 3-semesters)	Purdue Univ. FW

Professional Services & Outreach

PROFESSIONAL SERVICES

- 2025 **UrbanAI’ workshop SIGSPATIAL** Student Organizer
- 2025 **GeoAI’ workshop SIGSPATIAL** Reviewer
- 2025 **SIGSPATIAL** Reviewer
- 2024 **International Journal of Geographical Information Science** Reviewer
- 2024 **The Web Conference (WWW)**, Reviewer
- 2021-23 **ACM International Conference on Advances in Geographic Information Systems**, Reviewer
- 2022-24 **International Workshop on Health Intelligence on AAAI**, Reviewer
- 2022 **Conference on Information and Knowledge Management**, Reviewer

OUTREACH

2022-25	Spatial Women networking event for underrepresented groups , organizer	ACM SIGSPATIAL
2023	Spatial Women networking event for underrepresented groups , organizer	SIAM SDM
2022	Workshop on leveraging LLMs for Learning Geo-Entity Representation , co-organizer	UCGIS

LEADERSHIP ACTIVITIES

2022-now. **Spatial Women** General Chair

2022-24 **Computer Science Inclusivity, Diversity, Equity, and Advocacy (CS-IDEA)** Committee

Univ. of Minnesota

2022-23 **Computer Science Graduate Student Association (CSGSA)** President

Univ. of Minnesota

2021-22 **Computer Science Graduate Student Association (CSGSA)** Vice-president

Univ. of Minnesota

2017-18 **Club Seoul** President

Purdue univ. FW

MEDIA COVERAGE

2023 The David Rumsey Map Center e-newsletter (April 2023). “AI Advancements in Map Studies.” [Link]

Stanford Univ.

2022 Olivia Hultgren, UMN CSE News Page (October 2022). “University of Minnesota’s Knowledge Computing Lab turns location data into time-saving tools” [Link]

Univ. of Minnesota

Technical Skills

Programming PyTorch, Python, PostgreSQL, SQL, PySpark, Java, R

Tools QGIS, Elasticsearch, ArcGIS pro

References

Prof. Yao-Yi Chiang (Email: yaoyi@umn.edu)

Ph.D. Advisor, Associate Professor

Department of Computer Science and Engineering at University of Minnesota

Prof. Adolfo S. Coronado (Email: coronado@pfw.edu)

Chair of Computer Science and Associate

Department of Computer Science and Engineering at Purdue University Fort Wayne

Prof. Jin Soung Yoo (Email: yooj@pfw.edu)

M.Sc. Advisor, Professor

Department of Computer Science and Engineering at Purdue University Fort Wayne